NICHOLAS HO

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EDUCATION

University of Southern California Los Angeles, CA
Master of Science, Applied Data Science January 2026–Present (Expected May 2027)
Bachelor of Science, Quantitative Biology August 2021–December 2025

EXPERIENCE

Walmart Global Tech Sunnyvale, CA
Data Science Intern – Applied AI, Last Mile Delivery June 2026–Present

- Develop an anomaly-detection system for Walmart's Spark delivery platform to surface unusual driver behavior and prioritize potentially fraudulent cases for human review
- Extract and transform 180 days of driver-trip data from BigQuery using SQL, building driver-day behavioral profiles across the five largest Spark Zones
- Evaluated 50+ behavioral features using within-driver variability, between-driver variance, and one-way ANOVA, selecting 20+ signals for personalized anomaly detection
- Built and compared a statistical z-score framework and PCA-based Isolation Forest model, and applied prompt-engineering techniques to generate feature-level explanations and reviewer-friendly investigation rationales

USC Laboratory for Machine Learning, Health, and Biomedicine Los Angeles, CA
AI Research Intern – Professor Ruishan Liu May 2025–August 2025

- Wrote Python scripts to preprocess and standardize genomic sequences across formats (FASTA, BED, GZ), enabling uniform integration into a multi-task learning pipeline
- Authored Python tools to convert genomic coordinates from GRCh37 to GRCh38, ensuring compatibility across datasets and improving data alignments for downstream model training
- Conducted literature review and constructed a metadata schema to organize TCGA and other public datasets by predictive task categories, cutting down data retrieval time for model development by 30%

H&R Block Kansas City, MO
Software Engineering Intern – Virtual Tax Submission Team May 2024–August 2024

- Developed and deployed a digital consent collection tool leveraging Angular and C#, enhancing user experience for more than 1 million users by decreasing page reload friction
- Achieved 95% unit test coverage and supported compliant software delivery through Azure DevOps pipelines and CI/CD best practices
- Collaborated with product and legal teams to enhance data submission workflows, streamlining consent processes and mitigating user drop-off by ~20%

PROJECTS

CourtVision December 2025–Present
Python | scikit-learn | pandas | NumPy | React | Next.js

- Built a leakage-free season-over-season forecasting pipeline for 9 NBA per-game statistics using six seasons of player data; outperformed a repeat-last-season MAE baseline across all 9 targets on an unseen 2025-26 holdout
- Improved sample reliability by requiring 41 games and weighting training observations by games played, producing 361 qualified forecasts for the 2026-27 season
- Deployed an interactive React and Next.js website with searchable rankings, player comparisons, and complete model-performance and backtest reporting

Healthcare Pricing Prediction March 2025–December 2025
Python | matplotlib | Tableau | Jupyter Notebook

- Built an end-to-end regression pipeline in Python utilizing scikit-learn and XGBoost to predict hospital treatment costs, achieving an R^2 above 0.85
- Cleaned and preprocessed 50,000+ healthcare records in Jupyter Notebook by imputing missing values, encoding categorical variables, and scaling features, lowering model training time by 25% and boosting accuracy
- Assessed model accuracy using MAE, RMSE, and R^2, and visualized key pricing patterns with matplotlib and Tableau dashboards for stakeholder presentation

TECHNICAL SKILLS

Languages: Python, SQL, R, C++, C#, HTML/CSS, JavaScript
Machine Learning & AI: scikit-learn, XGBoost, Isolation Forest, PCA, prompt engineering, feature engineering, model evaluation
Data & Cloud: pandas, NumPy, BigQuery, Google Cloud Platform
Frameworks & Visualization: React, Angular, Flask, Tableau, matplotlib, seaborn
Developer Tools: Azure DevOps, Git, GitHub, Jira